Course Project
Spring 2025
COSC/ITEC-3506-002X: Software Engineering
Faculty of Computer Science and Technology
Algoma University
Total 24 Points

Term	25S
Total Points	24
Due Date	July 25, 2025

The project covers all the basic steps of Software Development Life Cycle (SDLC) with each phase corresponding to each step in the SDLC process.

Phase 1

Task: Topic Selection

Points: 0.5
Students are required to select a suitable topic or idea to develop into a working system. The topic selected must adhere to the following constraints:

- Should provide a viable solution to an existing real-world problem or requirement;
- If existing solutions are adapted, it must improve on existing works. Any cloned or copied repositories without significant feature revisions or modifications will be considered plagiarism;
- Must be a full-stack project, i.e. both backend and frontend, purely frontend or backend projects will not be accepted;
- Originality/uniqueness & practicality of the project will be taken into consideration during evaluation.

Choose one of the following project topic:
- Web3 based P2P chat application
- Web3 based P2P social media application
- Augmented reality (AR) based application for accessibility needs
- Augmented reality based application for social benefit
- Application for assisting cognitive disabilities.

Task 2: Product Vision

Points: 1

Create a product vision statement for the software using one of the following templates:

- https://www.prodpad.com/resources/free-product-vision-template
- https://learn.agileclassrooms.com/product-vision-templates

The product vision must be clear, concise, and practical.

Task 3: Development Model

Points: 1.5

Select a development methodology for your selected product such as Waterfall, Agile, Scrum, etc. Clear and logical explanations must be provided for the method chosen based on the project topic, group member expertise and roles, development timeline, and so on.

Resource: *https://www.geeksforgeeks.org/what-is-sdlc-model-and-its-phases*

Total Points: 3

Phase 2

Task 1: Software Architecture

Points: 3

Create a UML domain model and sequence diagrams for the project using LucidChart. Only the major workflows of the systems implementing the primary features need sequence diagrams. For example, for the event tracking system, only the event creation and overlap detection needs sequence diagrams.

Resource: https://www.youtube.com/watch?v=6XrL5jXmTwM&ab_channel=LucidSoftware

Total Points: 3

Phase 3

Task 1: Wireframing

Points: 2

Use LucidChart to design the wireframe for your project UI. Wireframes must be created for all planned pages/views of the project.

Resources:

- https://www.youtube.com/watch?v=qpH7-KFWZRI&ab_channel=CareerFoundry
- https://www.lucidchart.com/blog/how-to-make-a-wireframe-in-Lucidchart

Task 2: Design Documentation

Points: 3

Create a software design document using LucidChart. The design document must adhere to standard templates and must provide an accurate representation of the scope and plans for the project. Marking will be based on the overall quality and relevance of the document.

A list of functional and non-functional requirements must also be submitted along with the design document.

Resources:

- https://lucid.co/templates/software-design-document
- https://www.lucidchart.com/blog/how-to-create-software-design-documents

Total Points: 5

Phase 4

Task 1: Git Setup

Points: 1

Create a git repository for the project, invite all the project collaborators and the course instructor and upload the project template to the root branch of the repository. *Note: The instructor will monitor the commits periodically to ensure equal contribution and steady progress.*

Task 2: Software Development

Points: 8

Develop your proposed system. A functional system must be coded with all the proposed features. *If any initially proposed feature cannot be developed due to infeasibility, the team must provide a notice 2-weeks prior to the submission date to the course instructor for review and approval.*

Total Points: 9

Phase 5

Task 1: Project Testing

Points: 3

Perform functional and non-functional black-box testing of your software.
Resource: https://www.geeksforgeeks.org/software-testing-basics/

Task 2: Test Report

Points: 1

Create a simple test report with the executed test cases for the project both functional and non-functional. The test plan table should include the following fields: *Test No, Test Description, Assigned To, Expected Result, Actual Result, and Remarks*.

Total Points: 4

Phase 6

You must submit the following deliverables to BrightSpace. Your grade will be decided based on the accuracy, clarity, and authenticity of the following deliverables.

Task 1: Project Report

You must follow the following report format.
https://advice.writing.utoronto.ca/types-of-writing/lab-report/

Submit a report that includes all the details of the above mentioned phases. Include all relevant figures and references of each task.

Task 2: Source Code

Latest copy of the codebase must be zipped and submitted for review.

Task 3: Video Demo

A full demonstration of the project must be recorded in MP4 format and submitted to the instructor.